Outer Alignment & Moral Reasoning

An annotated bibliography and source-quality map for *The Trajectory of Outer Alignment in Large Language Models: Evaluating Normative Ethics, Pluralism, and Procedural Moral Reasoning (2025–2026)*

15 source families · 24 report references

Prepared 11 October 2026

The source base is useful—but uneven

The report's strongest evidence comes from process-focused benchmark papers and public datasets. Its weakest references are duplicate mirrors, profile pages, incomplete citations, and conceptual manuscripts used beyond their empirical scope.

> **Bottom line.** Treat MoReBench, Staircase, OMIT, MoralBench, CLASH, NormBank, Conscience Conflict, and AgentBreeder as the substantive core. Read "Are LLMs Bad at Moral Reasoning?" beside MoReBench, because it shows that elicitation and rubric design can reverse the apparent capability story.

What this bibliography does

Consolidates duplicates into source families; explains each work's contribution to the report; distinguishes evidence from framing; flags citation problems; and recommends the most defensible use of each source.

What "status" means

Core sources directly support major empirical claims. **Recommended** sources are useful within scope. **Use cautiously** marks under-review or conceptual work. **Verify** marks incomplete metadata or unresolved provenance.

Three recurring validity risks

Construct validity: does a benchmark measure moral competence or surface agreement? Elicitation validity: did the prompt expose the model's capability? Normative validity: whose values define the target?

One result is never enough

Outcome labels, reasoning rubrics, framing pairs, value profiles, stakeholder perspectives, sequential dilemmas, and scaffold-level tests answer different questions. Their scores should not be collapsed into one "alignment" number.

Recommended reading order

1. MoReBench, then Zhu and Lazar's reanalysis.
2. Staircase, OMIT, and CLASH for dynamics, framing, and perspective.
3. MoralBench and NormBank for descriptive value and norm profiles.
4. Positive Alignment and bidirectional alignment for the expanded design agenda.
5. AgentBreeder for system-level safety beyond base-model weights.

REPORT REFS 1 · PREPRINT / UNDER REVIEW

Use cautiously

Goal Multi-Stage Verification: A Defense-in-Depth Mathematical Framework for Mitigating Alignment in Deceptive Large Language Models

Anonymous authors. Under review at MathAI 2026. OpenReview manuscript.

Contribution. Proposes layered checks across outputs, internal signals, and formal reasoning to raise the cost of deceptive behavior. Its role in the report is mainly conceptual: it supplies notation and a defense-in-depth framing for the gap between intended reward, proxy reward, and a learned mesa-objective.

Assessment and best use. Relevant to inner alignment and verification, but only indirectly supports claims about normative outer alignment. The accessible manuscript is anonymous and under review; its reported mathematical-benchmark gains should not be treated as established evidence of deception detection. Best used to motivate a research direction, not as a settled result.

https://openreview.net/pdf?id=Co65Bj9zZ3

REPORT REFS 2 · PREPRINT / METADATA INCOMPLETE

Verify before citing

Evaluation-Conditioned Training: Teaching Models to Generalize Across Evaluators

OpenReview manuscript; bibliographic metadata were not recoverable from the report citation.

Contribution. The report uses this work to describe training across evaluators of different fidelity so that a model can learn how behavior should vary with oversight quality. That idea matters for outer alignment because ordinary human feedback is noisy, bounded, and vulnerable to proxy optimization.

Assessment and best use. The conceptual relevance is high, but the citation is incomplete and the source could not be independently inspected during this review. Recover the authors, venue, version date, and stable identifier before scholarly reuse. Also separate the technique’s intended use from the report’s speculative warning that it could be misused under weak oversight.

https://openreview.net/pdf/b664bb5dbffd96cece0ced9d7a7693481248f7b3.pdf

REPORT REFS 3 · POSITION PAPER / RESEARCH AGENDA

Recommended

Positive Alignment: Artificial Intelligence for Human Flourishing

Laukkonen, Ruben, et al. arXiv preprint, version 3, 2026.

Contribution. Defines positive alignment as safe, cooperative AI that also supports pluralistic, context-sensitive, user-authored human and ecological flourishing. It broadens the target from avoiding failures to creating beneficial attractors across data curation, training, evaluation, agents, and governance.

Assessment and best use. This is the report's clearest source for the move beyond harm reduction. It is strongest as a framework and agenda, not as proof that flourishing can already be operationalized without paternalism. Use it to structure design goals and open questions; pair it with empirical work when making claims about model behavior or intervention efficacy.

https://arxiv.org/html/2605.10310v3

REPORT REFS 4–6, 8, 11–12 · PRIMARY PAPER + DATASET FAMILY

Core source

MoReBench: Evaluating Procedural and Pluralistic Moral Reasoning in Language Models, More than Outcomes

Chiu, Yu Ying, et al. arXiv:2510.16380; ICLR 2026 paper, project commentary, and dataset mirrors.

Contribution. Introduces 1,000 moral scenarios with more than 23,000 expert-authored criteria and a 150-item theory subset covering five normative frameworks. It evaluates whether models identify considerations, weigh trade-offs, give useful recommendations, avoid harm, and reason within specified moral traditions.

Assessment and best use. This is the report's central empirical source. Its process focus and pluralistic theory subset directly support the report's argument that outcome-only tests are insufficient. The six listed references are not six independent studies: they include the paper, lab commentary, mirrors, and dataset hosting. Cite the primary paper once, then cite the dataset separately only when data access matters.

https://arxiv.org/abs/2510.16380

REPORT REFS 7 · PEER-REVIEWED CONFERENCE PAPER

Core source

The Staircase of Ethics: Probing LLM Value Priorities through Multi-Step Induction to Complex Moral Dilemmas

Wu, Ya, et al. Proceedings of EMNLP 2025, pp. 15950–15970.

Contribution. Presents 3,302 five-stage moral dilemmas and evaluates how model value priorities change as conflicts accumulate. The study reports shifting and sometimes non-transitive preferences, especially as care, fairness, liberty, loyalty, authority, and sanctity come into tension.

Assessment and best use. A strong source for the claim that static, one-shot dilemmas miss path dependence. Its generated scenarios and value mappings are still measurement choices, so preference shifts should not automatically be equated with deep internal incoherence. Use it to justify sequential stress tests, while distinguishing observed response instability from claims about a model's underlying moral architecture.

https://aclanthology.org/2025.emnlp-main.806.pdf

REPORT REFS 9 · POSITION PAPER / SYSTEMATIC REVIEW

Recommended with scope note

Position: Towards Bidirectional Human-AI Alignment

Shen, Hua, et al. NeurIPS 2025 position paper.

Contribution. Reviews more than 400 papers and argues that alignment is not only AI adaptation to human values. It also includes human cognitive, behavioral, and societal adaptation to AI, with attention to agency, long-term interaction, value modeling, and mutual understanding.

Assessment and best use. Useful for expanding the unit of analysis from model outputs to reciprocal human-AI systems. It is a framing paper rather than a benchmark, and "aligning humans to AI" can be normatively hazardous if it becomes coercive adaptation. Cite it for the bidirectional framework and research gaps, not as empirical proof that reciprocal adaptation improves moral alignment.

https://Openreview.net/pdf?id=HcTiacDN8N

REPORT REFS 10 · PREPRINT BENCHMARK PAPER

Core source

OMIT the Action: Measuring Framing-Invariant Omission Bias under Philosophical Disagreement

Lee, Sihyeon, et al. arXiv:2610.07847, 2026.

Contribution. Builds 218 paired-frame scenarios across ten conflict types using five philosophical perspectives. A model displays omission bias when it recommends agent-level inaction in both frames even though the physical outcome reverses. The study also evaluates inference-time prompts intended to reduce framing sensitivity.

Assessment and best use. Directly supports the report's discussion of action-versus-omission framing. The report attributes the method to Cheung et al. (2025), while this cited source names Lee et al. (2026) and treats Cheung et al. as prior work; that attribution should be corrected. Reduced omission-bias rate is not automatically better if an intervention merely creates action bias, so inspect the full response distribution.

https://arxiv.org/html/2610.07847v1

REPORT REFS 14 · CRITICAL REANALYSIS / PREPRINT

Core counterpoint

Are LLMs Bad at Moral Reasoning?

Zhu, Menghang, and Seth Lazar. arXiv:2606.11635, 2026.

Contribution. Reuses MoReBench to test whether apparent underperformance reflects the models' competence or the evaluation setup. When models generate rubrics rather than answer open-endedly, the strongest systems recover most human-identified considerations and add many plausible considerations omitted by the human rubrics.

Assessment and best use. Essential because it shows that benchmark conclusions depend on elicitation and rubric coverage. It does not invalidate MoReBench; it changes the interpretation of low scores. Read the two together: one documents process-evaluation gaps, the other shows that the evaluator can under-recognize capability. Neither alone establishes dependable moral agency in deployment.

https://arxiv.org/pdf/2606.11635

REPORT REFS 15, 17 · BENCHMARK PREPRINT

Recommended

MoralBench: Moral Evaluation of LLMs

Ji, Jianchao, et al. arXiv:2406.04428, revised version.

Contribution. Adapts Moral Foundations Theory to compare model profiles across care, fairness, loyalty, authority, sanctity, and liberty. The benchmark emphasizes moral identity, contextual sensitivity, and consistency rather than only whether an answer matches a single moral label.

Assessment and best use. Useful for profiling differences across models and releases, but it measures responses through one moral-psychology framework rather than moral truth. The report's very specific cross-model statistics should be checked against the paper's tables before reuse. References 15 and 17 are duplicate routes to the same study and should be consolidated in a formal bibliography.

https://arxiv.org/html/2406.04428v2

REPORT REFS 16 · REPOSITORY MANUSCRIPT / INCOMPLETE CITATION

Verify before citing

A Vignette-Based Inquiry into Moral Reasoning Across Six Large Language Models

Institutional-repository item; author, year, and venue are missing from the report entry.

Contribution. The report lists this as support for comparative, vignette-based evaluation across six models. The title suggests controlled moral scenarios intended to compare reasoning patterns rather than broad capability scores.

Assessment and best use. The reference is not integrated clearly into the report's argument and its printed URL is line-broken. Do not rely on it until the full metadata and stable landing page are recovered. If retained, specify which claim it supports, the models tested, how vignettes were validated, and whether the work evaluates outcomes, rationales, or both.

No verified stable URL included.

REPORT REFS 18 · BENCHMARK PREPRINT + DATASET

Core source

CLASH: Evaluating Language Models on Judging High-Stakes Dilemmas from Multiple Perspectives

Lee, Ayoung, Ryan Sungmo Kwon, Peter Railton, and Lu Wang. arXiv:2504.10823, revised version.

Contribution. Introduces 345 long-form high-stakes dilemmas and 3,795 character perspectives. It tests ambivalence, psychological discomfort, value shifts, and steerability between competing values; strong models perform much worse on ambivalent and dynamically shifting cases than on clear-cut cases.

Assessment and best use. A strong complement to theory-based benchmarks because it centers stakeholder perspective and temporal value change. It tests whether a model can represent a character's values, not whether those values are ethically correct. Use it for pluralistic perspective-taking and steerability claims, while avoiding the inference that fidelity to a perspective equals moral alignment.

https://arxiv.org/html/2504.10823v2

REPORT REFS 19 · WORKSHOP PAPER

Useful qualitative complement

Conscience Conflict? Evaluating Language Models' Moral Understanding

Hota, Asutosh, and Jussi P. P. Jokinen. MoMLeT 2025 workshop proceedings.

Contribution. Evaluates five open-source models on six human-authored, high-stakes dilemmas that require both a decision and a justification. Responses are coded by ethical framework and assessed for coherence, empathy, context sensitivity, and reasoning depth.

Assessment and best use. The qualitative design exposes rationale shifts that binary tests can hide. Its evidence base is deliberately small, so it should not carry broad prevalence claims across model families. Use it to motivate close reading, prompt-sensitivity audits, and mixed-method evaluation; pair it with larger benchmarks for estimates of frequency or comparative performance.

https://ceur-ws.org/Vol-4004/paper3.pdf

REPORT REFS 20, 22 · PEER-REVIEWED PAPER + DATASET

Core descriptive-norm source

NormBank: A Knowledge Bank of Situational Social Norms

Ziems, Caleb, et al. ACL 2023, pp. 7756–7776; public data and code repository.

Contribution. Provides 155,000 situational norms grounded in settings, social roles, attributes, and cultural or physical constraints. The resource is designed to model how small context changes can cancel or reverse a norm, and includes a public implementation repository.

Assessment and best use. Excellent for contextual social-norm inference and reproducible dataset work. It is descriptive: annotator judgments and covered cultures do not constitute a universal normative standard. The report sometimes blends NormBank with Delphi; keep the dataset, the downstream moral-judgment model, and later Delphi evaluations bibliographically distinct. References 20 and 22 should be cited as repository and paper, respectively.

https://aclanthology.org/2023.acl-long.429/

REPORT REFS 21 · TEACHING NOTES / SECONDARY SOURCE

Context only

What Do Moral Benchmarks Measure? — course lecture notes

Stanford course material; author and publication date not stated on the cited page.

Contribution. Uses Delphi and NormBank to ask what moral benchmarks assume about agency, whether they measure the intended construct, and how cultural coverage and uncertainty could be improved. It also points readers to primary research on perceived moral expertise.

Assessment and best use. Contrary to the report's terse reference, the page is relevant rather than a generic database lecture. Still, it is course material, not a primary research publication. Use it as a critical-reading guide or to locate cited papers, not as the evidentiary basis for empirical results. Replace its reported numbers with citations to the underlying Delphi or NormBank publications when possible.

https://web.stanford.edu/class/cs186/internal/lectures/representation_2.html?print-pdf

REPORT REFS 23 · PEER-REVIEWED CONFERENCE PAPER

Core agentic-safety source

AgentBreeder: Mitigating the AI Safety Impact of Multi-Agent Scaffolds via Self-Improvement

Rosser, J., and Jakob Foerster. ICLR 2025; arXiv:2502.00757.

Contribution. Uses multi-objective evolutionary search to discover multi-agent scaffolds under capability and safety objectives. Defensive search improves safety while preserving capability, whereas adversarial search finds scaffolds that remain capable yet amplify safety weaknesses.

Assessment and best use. The strongest source in the list for the claim that alignment belongs to the entire scaffold, not just the base model. Its benchmarks address safety and task performance rather than normative moral reasoning, so the report's connection to "thick value models" is an extrapolation. Use it to justify system-level red teaming and joint optimization, with that scope made explicit.

https://arxiv.org/abs/2502.00757

REPORT REFS 13, 24 · DISCOVERY AIDS

Do not use as evidence

Author and researcher profile pages

Sydney Levine on alphaXiv; Haoran Ye on Google Scholar.

Contribution. These entries point to researcher profiles rather than specific studies. They may help locate related work, citation records, or author histories, but they do not independently substantiate a claim in the report.

Assessment and best use. Remove them from a formal works-cited list unless the profiles themselves are the object of discussion. Replace each with the exact paper that supports the associated claim. Their inclusion inflates the apparent source count and makes it harder to distinguish primary evidence from navigation pages.

No verified stable URL included.

No single benchmark establishes moral alignment

The literature becomes coherent when each source is assigned to the layer it can actually measure. The resulting evaluation stack is multi-method by design.

Layer	Best sources here	Question answered
Normative target	Positive Alignment; Bidirectional Human-AI Alignment	What should the system promote, and who gets to define it?
Procedural competence	MoReBench; Zhu & Lazar	Can the model identify and weigh considerations—and does the evaluator recognize them?
Value profile	MoralBench	Which moral foundations appear stable across prompts, models, and releases?
Sequential stability	Staircase of Ethics	Do value priorities remain coherent as stakes and context accumulate?
Framing robustness	OMIT	Does action-versus-omission framing change the recommendation while outcomes reverse?
Perspective pluralism	CLASH	Can the model represent ambivalence, discomfort, and changing stakeholder values?
Rationale coherence	Conscience Conflict	Are justifications principled and consistent, not merely socially acceptable?
Contextual norms	NormBank	Can the system model norms that depend on roles, settings, and culture?
System-level safety	AgentBreeder	Does the scaffold preserve safety when multiple agents and feedback loops interact?

Practical implication. A deployment claim should report a profile, not a scalar: outcome quality, reasoning coverage, frame consistency, sequential stability, stakeholder sensitivity, uncertainty calibration, and scaffold-level safety.

Six fixes make the evidence base reusable

The report points to a strong literature, but its works-cited list needs consolidation and metadata repair before it can function as a scholarly bibliography.

Collapse duplicates

Combine mirrors and landing pages into one citation per work. Keep a separate dataset citation only when data access or versioning matters.

Replace profiles

Author profiles are discovery aids, not evidence. Replace references 13 and 24 with the exact papers meant to support the surrounding claims.

Recover metadata

Complete the author, date, venue, and stable identifier for Evaluation-Conditioned Training and the six-model vignette study.

Correct attribution

The report attributes the cited omission-bias method to Cheung et al.; reference 10 is a 2026 paper by Lee et al. that treats Cheung et al. as prior work.

Recheck exact numbers

Verify every model-comparison statistic against the primary table and version. Secondary summaries are useful for orientation, not numerical authority.

Label publication status

Distinguish peer-reviewed proceedings, preprints, under-review manuscripts, repositories, and course notes. They carry different evidentiary weight.

> **Best reuse pattern.** Cite the primary study for claims, the dataset for replication, and a secondary explainer only for interpretation. Do not count those as independent confirmations.

All 24 report references accounted for

The report prints 24 numbered references, but several point to the same work or to a non-evidentiary profile. This map preserves the original numbering while showing the recommended bibliographic treatment.

Report no.	Source family	Recommended treatment
1	Defense-in-depth manuscript	Primary manuscript; under review
2	Evaluation-Conditioned Training	Incomplete metadata; verify
3	Positive Alignment	Primary position paper
4–6	MoReBench paper + commentary	Consolidate around primary paper
7	Staircase of Ethics	Primary conference paper
8	MoReBench alphaXiv mirror	Duplicate / commentary
9	Bidirectional Human-AI Alignment	Primary position paper
10	OMIT	Primary preprint; correct attribution
11	MoReBench Hugging Face paper page	Duplicate discovery page
12	MoReBench dataset	Dataset citation
13	Sydney Levine profile	Discovery aid only

Reference audit, continued

Report no.	Source family	Recommended treatment
14	Are LLMs Bad at Moral Reasoning?	Primary counteranalysis
15, 17	MoralBench	Duplicate routes to one paper
16	Six-model vignette inquiry	Incomplete citation; verify
18	CLASH	Primary benchmark paper
19	Conscience Conflict	Primary workshop paper
20	NormBank repository	Code/data citation
21	Stanford moral-benchmarks lecture	Secondary teaching notes
22	NormBank ACL paper	Primary peer-reviewed paper
23	AgentBreeder	Primary conference paper
24	Haoran Ye Google Scholar profile	Discovery aid only

URLs are included only where a stable address was printed in the source report or returned by a public source check. A missing link is a deliberate provenance safeguard, not an indication that the work does not exist.